

Does Thinking More always Help?

Understanding Test Time Scaling In Reasoning Models

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1 Test-time Scaling Capability in LLMs for Reasoning

- our empirical investigation uncovers a nuanced phenomenon: extending thinking at test-time initially boosts model accuracy, but performance degrades subsequently with prolonged thinking
1. Why does additional thinking beyond a certain point degrade the model performance?
 2. analyze a simple one-dimensional probabilistic framework
 3. empirically assess the variance of reasoning-model outputs under extended thinking by measuring the entropy of their output distributions.
 4. fix: Instead of continuing one thinking trace, we allocate the same token budget across multiple independent thinking paths and select the final answer via majority voting.

2 Analysis of Test-Time Scaling in SoTA Reasoning LLMs

Thinking process

$$x \rightarrow z \rightarrow y$$

3 Problems

4 References